

## MOGILEV, Belarus

WMO: 268630

Lat: 53.956N

Lon: 30.096E

Elev: 632

StdP: 14.36

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB  |             | Humidification DP/MCDB and HR |            |             |             |            |             | Coldest Month WS/MCDB |             |             |             | MCWS/PCWD to 99.6% DB |           | WSF          |
|---------------|-------------|-------------|-------------------------------|------------|-------------|-------------|------------|-------------|-----------------------|-------------|-------------|-------------|-----------------------|-----------|--------------|
|               |             |             | 99.6%                         |            |             | 99%         |            |             | 0.4%                  |             | 1%          |             |                       |           |              |
|               | 99.6%       | 99%         | DP                            | HR         | MCDB        | DP          | HR         | MCDB        | WS                    | MCDB        | WS          | MCDB        | MCWS                  | PCWD      |              |
| (a)<br>1      | (b)<br>-7.3 | (c)<br>-1.3 | (d)<br>-11.2                  | (e)<br>3.1 | (f)<br>-6.3 | (g)<br>-5.7 | (h)<br>4.1 | (i)<br>-0.9 | (j)<br>25.7           | (k)<br>30.6 | (l)<br>23.2 | (m)<br>32.9 | (n)<br>5.0            | (o)<br>10 | (p)<br>0.597 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month<br>DB Range | Cooling DB/MCWB |             |             |             |             |             | Evaporation WB/MCDB |             |             |             |             |             | MCWS/PCWD<br>to 0.4% DB |            |
|---------------|---------------------------|-----------------|-------------|-------------|-------------|-------------|-------------|---------------------|-------------|-------------|-------------|-------------|-------------|-------------------------|------------|
|               |                           | 0.4%            |             | 1%          |             | 2%          |             | 0.4%                |             | 1%          |             | 2%          |             |                         |            |
|               |                           | DB              | MCWB        | DB          | MCWB        | DB          | MCWB        | WB                  | MCDB        | WB          | MCDB        | WB          | MCDB        | MCWS                    | PCWD       |
| (a)<br>7      | (b)<br>18.1               | (c)<br>84.4     | (d)<br>67.9 | (e)<br>80.9 | (f)<br>66.2 | (g)<br>78.0 | (h)<br>65.1 | (i)<br>69.9         | (j)<br>80.8 | (k)<br>68.1 | (l)<br>78.0 | (m)<br>66.5 | (n)<br>75.6 | (o)<br>8.1              | (p)<br>180 |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 66.2                            | 98.9 | 75.1 | 64.5 | 92.9 | 72.7 | 62.8 | 87.6 | 70.9 | 34.2          | 81.3 | 32.7 | 78.2 | 31.4 | 75.8 | 79.3              |

## Extreme Annual Design Conditions

| Extreme Annual WS |          |          | Extreme Annual Temperature |          |                    |         | n-Year Return Period Values of Extreme Temperature |          |            |          |            |          |            |          |    |    |
|-------------------|----------|----------|----------------------------|----------|--------------------|---------|----------------------------------------------------|----------|------------|----------|------------|----------|------------|----------|----|----|
|                   |          |          | Mean                       |          | Standard Deviation |         | n=5 years                                          |          | n=10 years |          | n=20 years |          | n=50 years |          |    |    |
| 1%                | 2.5%     | 5%       | Min                        | Max      | Min                | Max     | Min                                                | Max      | Min        | Max      | Min        | Max      | Min        | Max      | DB | WB |
| (a) 21.5          | (b) 19.2 | (c) 17.2 | (e) -14.0                  | (f) 89.0 | (g) 5.8            | (h) 3.8 | (i) -18.1                                          | (j) 91.8 | (k) -21.5  | (l) 94.0 | (m) -24.7  | (n) 96.2 | (o) -28.9  | (p) 99.0 |    |    |
| (a) 21.5          | (b) 19.2 | (c) 17.2 | (e) -14.3                  | (f) 73.2 | (g) 5.7            | (h) 2.4 | (i) -18.4                                          | (j) 75.0 | (k) -21.7  | (l) 76.4 | (m) -24.9  | (n) 77.8 | (o) -29.1  | (p) 79.6 |    |    |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 43.8   | 21.9  | 22.9  | 31.5  | 45.4  | 55.6  | 61.5  | 65.9  | 63.6  | 53.9  | 43.0  | 33.6  | 25.2  |
|                                                                               | DBStd       | 17.78  | 11.11 | 11.44 | 8.64  | 7.85  | 8.01  | 6.04  | 5.43  | 6.28  | 6.40  | 7.52  | 8.75  | 10.58 |
|                                                                               | HDD50       | 3957   | 871   | 758   | 573   | 180   | 36    | 1     | 0     | 0     | 35    | 240   | 493   | 770   |
|                                                                               | HDD65       | 7937   | 1336  | 1178  | 1037  | 587   | 306   | 138   | 55    | 104   | 336   | 683   | 942   | 1235  |
|                                                                               | CDD50       | 1687   | 0     | 0     | 1     | 43    | 209   | 347   | 492   | 422   | 151   | 22    | 0     | 0     |
|                                                                               | CDD65       | 192    | 0     | 0     | 0     | 0     | 15    | 33    | 82    | 60    | 2     | 0     | 0     | 0     |
|                                                                               | CDH74       | 1712   | 0     | 0     | 0     | 6     | 141   | 261   | 713   | 557   | 34    | 0     | 0     | 0     |
|                                                                               | CDH80       | 400    | 0     | 0     | 0     | 1     | 17    | 33    | 181   | 165   | 3     | 0     | 0     | 0     |
| Wind                                                                          | WSAvg       | 8.4    | 9.4   | 9.1   | 9.4   | 8.6   | 8.2   | 7.6   | 6.8   | 7.0   | 7.3   | 8.4   | 9.1   | 9.7   |
| Precipitation                                                                 | PrecAvg     | 24.6   | 1.6   | 1.5   | 1.5   | 1.7   | 2.3   | 2.6   | 3.5   | 2.5   | 2.0   | 2.2   | 1.8   | 1.5   |
|                                                                               | PrecMax     | 35.9   | 3.8   | 2.5   | 2.9   | 4.5   | 4.8   | 5.4   | 8.0   | 5.8   | 4.1   | 5.6   | 3.7   | 2.6   |
|                                                                               | PrecMin     | 18.5   | 0.4   | 0.7   | 0.2   | 0.4   | 0.4   | 1.0   | 0.7   | 0.1   | 0.6   | 0.2   | 0.1   | 0.7   |
|                                                                               | PrecStd     | 4.1    | 0.7   | 0.5   | 0.7   | 0.9   | 1.1   | 1.0   | 1.9   | 1.4   | 1.0   | 1.4   | 0.9   | 0.6   |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 42.6  | 43.8  | 58.6  | 73.5  | 82.4  | 84.1  | 89.1  | 91.3  | 78.6  | 67.5  | 53.2  | 46.7  |
|                                                                               |             | MCWB   | 41.2  | 40.5  | 47.1  | 55.8  | 65.9  | 68.0  | 68.6  | 68.8  | 61.8  | 58.5  | 50.6  | 45.2  |
|                                                                               | 2%          | DB     | 38.6  | 39.5  | 51.8  | 68.3  | 78.4  | 80.4  | 85.0  | 84.2  | 73.1  | 60.6  | 49.3  | 42.8  |
|                                                                               |             | MCWB   | 37.1  | 37.3  | 43.7  | 53.8  | 63.4  | 66.0  | 68.4  | 67.4  | 60.4  | 54.4  | 47.2  | 41.8  |
|                                                                               | 5%          | DB     | 36.5  | 37.2  | 46.6  | 63.6  | 73.7  | 77.1  | 81.1  | 80.0  | 68.3  | 56.8  | 46.4  | 39.7  |
|                                                                               |             | MCWB   | 35.3  | 35.7  | 41.1  | 51.9  | 60.9  | 64.9  | 67.0  | 65.9  | 58.7  | 52.8  | 44.7  | 38.7  |
|                                                                               | 10%         | DB     | 34.1  | 35.3  | 42.6  | 59.2  | 69.8  | 73.7  | 77.8  | 76.0  | 64.6  | 53.6  | 44.2  | 36.5  |
|                                                                               |             | MCWB   | 33.2  | 34.0  | 38.8  | 49.5  | 58.5  | 62.7  | 65.7  | 64.0  | 56.7  | 50.3  | 42.9  | 35.3  |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 41.3  | 41.0  | 48.9  | 58.3  | 68.2  | 70.6  | 72.7  | 71.7  | 64.9  | 60.4  | 51.3  | 45.5  |
|                                                                               |             | MCDB   | 42.5  | 43.4  | 56.2  | 68.6  | 79.0  | 79.1  | 83.9  | 83.2  | 74.5  | 64.1  | 52.8  | 46.1  |
|                                                                               | 2%          | WB     | 37.2  | 37.7  | 44.5  | 55.0  | 65.2  | 68.1  | 70.4  | 69.4  | 62.1  | 55.7  | 47.7  | 41.8  |
|                                                                               |             | MCDB   | 38.0  | 39.0  | 50.3  | 65.1  | 75.5  | 77.3  | 81.4  | 81.1  | 69.6  | 59.3  | 48.9  | 42.5  |
|                                                                               | 5%          | WB     | 35.2  | 35.8  | 41.5  | 53.0  | 62.4  | 66.2  | 68.3  | 67.5  | 59.8  | 53.0  | 44.8  | 38.8  |
|                                                                               |             | MCDB   | 36.2  | 36.9  | 45.7  | 61.8  | 71.7  | 75.1  | 77.9  | 77.8  | 66.0  | 56.3  | 46.0  | 39.7  |
|                                                                               | 10%         | WB     | 33.3  | 34.0  | 39.1  | 50.7  | 60.1  | 64.2  | 66.6  | 65.4  | 57.8  | 50.8  | 42.8  | 35.4  |
|                                                                               |             | MCDB   | 34.0  | 34.9  | 42.3  | 57.8  | 67.8  | 71.8  | 75.6  | 73.6  | 63.1  | 53.4  | 43.9  | 36.2  |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 8.1   | 9.7   | 12.3  | 17.1  | 18.3  | 17.6  | 18.1  | 18.7  | 16.6  | 12.5  | 7.3   | 7.3   |
|                                                                               |             | MCDBR  | 6.3   | 8.1   | 18.0  | 24.1  | 23.7  | 22.1  | 22.8  | 24.3  | 22.7  | 17.3  | 9.6   | 7.1   |
|                                                                               |             | MCWBR  | 6.2   | 6.9   | 11.7  | 12.6  | 11.8  | 11.1  | 9.7   | 10.6  | 12.4  | 11.8  | 8.0   | 6.8   |
|                                                                               | 5% WB       | MCDBR  | 6.0   | 7.7   | 15.9  | 21.4  | 21.7  | 20.0  | 20.6  | 21.8  | 19.3  | 14.7  | 8.5   | 7.0   |
| MCWBR                                                                         |             | 5.9    | 7.0   | 11.1  | 12.3  | 11.8  | 11.5  | 10.0  | 10.7  | 11.9  | 10.9  | 7.5   | 7.0   |       |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.290  | 0.323 | 0.353 | 0.410 | 0.398 | 0.381 | 0.409 | 0.418 | 0.376 | 0.350 | 0.330 | 0.293 |       |
|                                                                               | taud        | 2.145  | 2.091 | 2.135 | 2.173 | 2.289 | 2.360 | 2.297 | 2.264 | 2.343 | 2.387 | 2.262 | 2.179 |       |
|                                                                               | Ebn at Noon | 205    | 236   | 257   | 258   | 269   | 275   | 264   | 253   | 250   | 230   | 189   | 178   |       |
|                                                                               | Edh at Noon | 21     | 31    | 37    | 41    | 39    | 36    | 38    | 37    | 30    | 23    | 19    | 17    |       |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 219    | 493   | 930   | 1284  | 1667  | 1809  | 1727  | 1437  | 958   | 480   | 173   | 128   |       |
|                                                                               | RadStd      | 46     | 93    | 139   | 148   | 142   | 153   | 160   | 139   | 83    | 95    | 31    | 36    |       |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only            |  | +0.94 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (11 neighbors) |  | +1.06 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | -320  | +180  | N/A   |

Nomenclature: See separate page